Lindenfeld Household Travel Survey 2027

Methods report

Lindenfeld Transport Office · Version 1.1 · March 2028

Synthetic sample: the town, the survey and all figures in this report are invented.

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1 Purpose of the survey

The Household Travel Survey describes how residents of Lindenfeld travel on an ordinary day: how many trips they make, for what purpose, how far they go and which means of transport they use. The results feed the municipal transport plan, the annual cycling report and the forecasts for the regional bus network.

The 2027 survey is the third in the series. It repeats the core questions of the earlier rounds so that trends can be followed, and adds questions on working from home and on the use of shared e-scooters.[1]

This report documents how the survey was carried out. It is written for analysts who use the data and for readers who want to judge how far the published figures can be trusted. The results themselves are published separately.

1.1 Who commissioned the survey

The survey was commissioned by the Lindenfeld Transport Office and funded jointly with the regional transport authority. Fieldwork was carried out by an external research contractor under a framework agreement; the Transport Office kept responsibility for the questionnaire, the weighting and the publication of results.

[1] The earlier rounds took place in 2017 and 2022. The 2022 questionnaire was shortened because of the pandemic, so some of its results are not directly comparable.

2 Sampling and fieldwork

2.1 Sample design

A random sample of 5,930 addresses was drawn from the municipal address register, stratified by the twelve town districts. Districts with fewer residents were oversampled so that each district has at least 300 responding households.[2]

Each household was assigned a travel day between Tuesday and Thursday. Mondays, Fridays and public holidays were excluded because travel on those days differs from the typical working week.

2.2 Fieldwork

Fieldwork ran from 12 April to 30 June 2027.[3] Invited households received a letter with a personal code for the online questionnaire. Paper questionnaires were sent to households that had not responded after one reminder.

A telephone helpline was open on weekdays. Interviews by telephone were offered to residents who could not complete the questionnaire online or on paper.

[2] Of the 5,930 invited households, 1,862 completed the questionnaire, a response rate of 31.4 %.

[3] Fieldwork was paused from 22 to 26 May because of the regional holiday week.

3 Questionnaire

The questionnaire has three parts: a household part, a person part for every member aged six or over, and a trip diary for the assigned travel day. The household part is completed once; the person part and the trip diary are completed for each member.

Most questions were kept from 2022. New questions cover the number of days worked from home in a typical week and the use of shared e-scooters and bicycles.[4] ~~Questions on car ownership were moved to the end of the questionnaire to reduce drop-outs.~~ The trip diary itself was not changed.

3.1 Pre-test

A pre-test with 120 households in two districts was carried out in February 2027. It led to shorter instructions for the trip diary and to a clearer definition of a trip that is interrupted for a short stop.

[4] The full questionnaire, in German and English, is published together with the data set.

4 Weighting and estimation

Responses are weighted so that the sample matches the population of Lindenfeld by district, household size, age group and sex.[5] Weights are first computed for households and then adjusted for persons.

Weights were calibrated by raking over the four margins. Weights larger than five times the mean weight were trimmed, and the remaining weights were rescaled to the population total.

4.1 Estimates and margins of error

All published shares and averages are weighted estimates. Margins of error are given at the 95 % level and take the stratified design into account. Figures based on fewer than 50 respondents are marked and should be read with caution.

[5] Population totals are taken from the municipal register on 31 December 2026.

5 Data quality

Completed questionnaires were checked for missing answers, implausible trip times and trips that do not connect, for example a trip that ends in a different place from where the next one starts.[6] Inconsistent trips were corrected where the answer was clear from the rest of the diary; otherwise they were flagged.

Households that reported no trips at all on the travel day were kept in the data. Their share is similar to that in the 2022 survey.[7]

5.1 Item non-response

Item non-response is low for most questions. The question on household income was answered by 78 % of households; for the analysis, missing incomes were imputed from household size, district and employment status.

[6] About 4 % of all reported trips needed a correction of this kind.

[7] Diaries without any trip on the travel day were checked by telephone with a sample of these households.

Annex A Definitions

Trip: a movement from one place to another for a single purpose. A trip that includes a stop of less than five minutes, for example to drop off a child, counts as one trip.

Main mode: the means of transport used for the longest distance of a trip.

Household: all people who live at the same address and share living costs.

Travel day: the day for which the household reports its trips; one of Tuesday, Wednesday or Thursday.

Active modes: walking and cycling, including e-bikes with assistance up to 25 km/h.